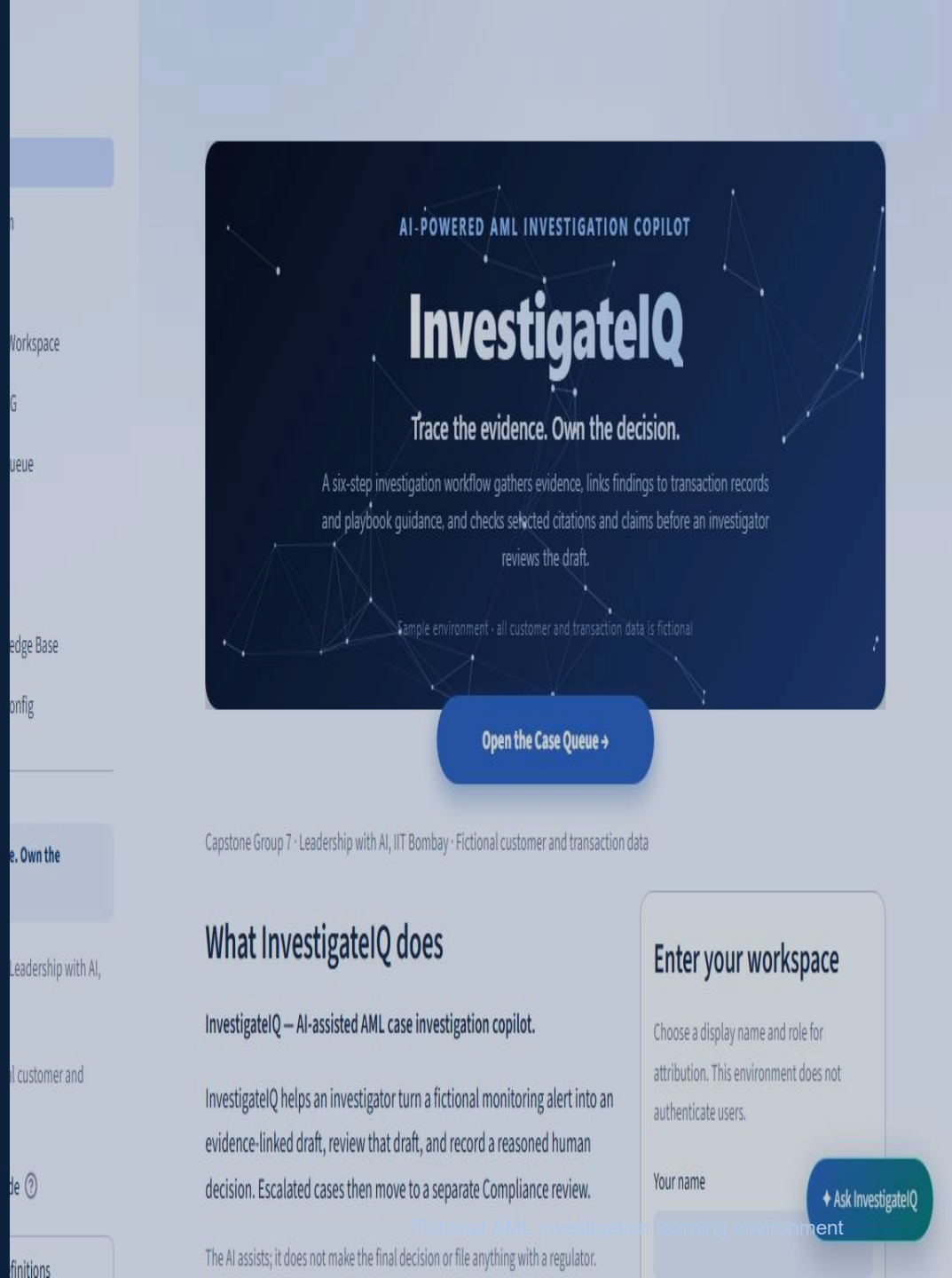
CAPSTONE GROUP 7

InvestigateIQ

Trace the evidence. Own the decision.

A transparent, AI-assisted workflow for fictional AML case investigations.

Prototype- https://investigateiq-app-production.up.railway.app/



Soumya — presenter | Laxman — backup

5:35 core slides • live demo and questions within the 15-minute session

PROGRAM STATEMENT

What this capstone will deliver—and what it will not

Design and demonstrate a fictional AML review workflow where every report claim is traceable to supplied material, limitations stay visible, and workflow actions remain named human decisions.

Track

Blueprint + Prototype + Detailed code + Final results

Measures

Traceability, visible gaps, clear states, and required rationale

Non-goals

No real data, autonomous decision, legal conclusion, or external filing

Team: Rahul Chainani · Pankaj Bharsakale · Lakshmi Dudgikar · Rahul Sharma · Laxman Singh · Soumya · Sweta Singh

Soumya presents the session; Laxman is the backup presenter.

THE PRODUCT

From a stored alert to an accountable human action

1

Queue

Find and filter the fictional alert

2

Workspace

Build or replay the review

3

Evidence

Read exact records and chunks

4

Decision

Record a required rationale

5

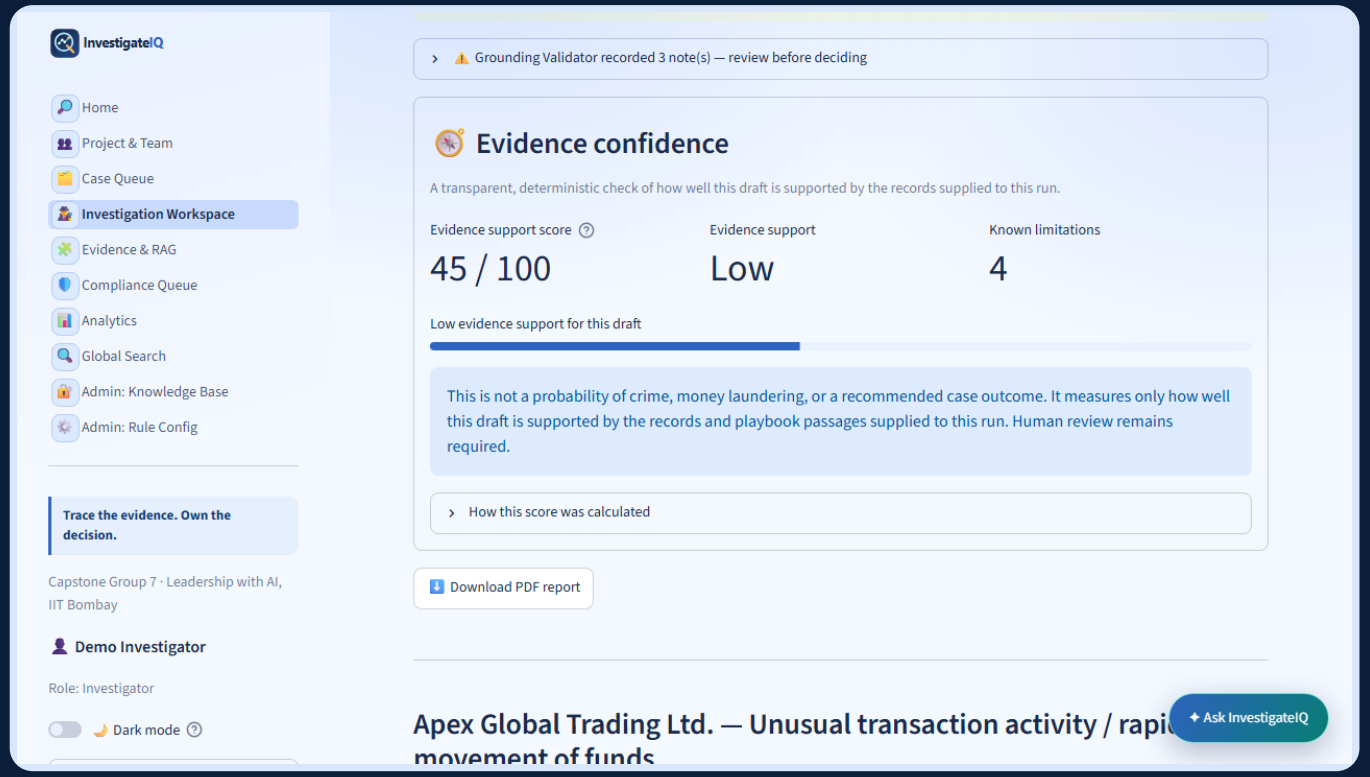
Compliance

Perform a second human review

The application supports investigation. It never closes, escalates, refers, or files an action automatically.

RESPONSIBLE AI

Evidence confidence exposes support gaps—it does not predict wrongdoing

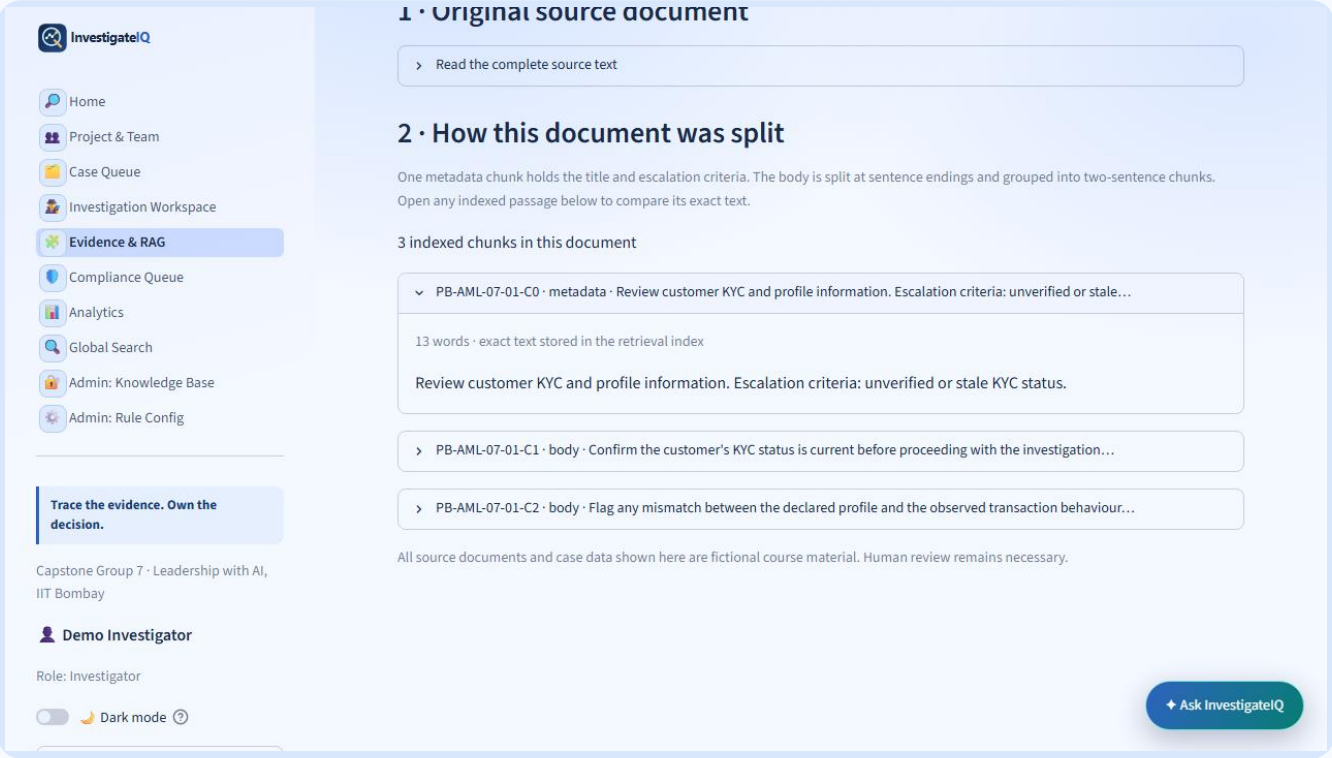


Coverage — Supplied records and guidance that support the draft

Limitations — Known missing or unresolved evidence

Boundary — Not a crime, legality, or outcome probability

The coding logic keeps the source trail inspectable



1. **OKF source**
   ID, scenario, title, criteria, body

2. **Chunk builder**
   C0 metadata + two-sentence body chunks

3. **FTS5 + BM25**
   Local keyword match and rank

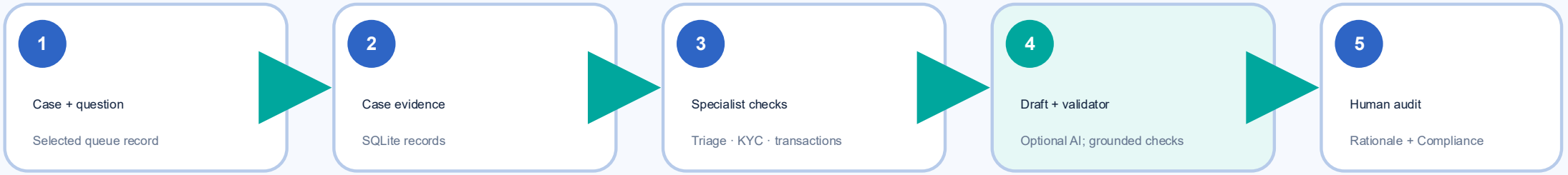
4. **RAG context**
   Retrieved IDs produce a cited, reviewable answer

OKF source → chunk builder → FTS5/BM25 → retrieved IDs → cited response

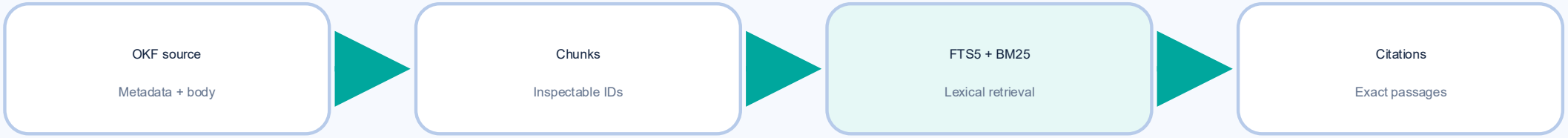
INVESTIGATEIQ DATA FLOW

Evidence first. Human decision last.

CASE-REVIEW FLOW



KNOWLEDGE-RETRIEVAL FLOW



Control boundary: retrieval finds supplied text; validation checks selected claims; confidence exposes limitations; no action is automated.

A named person interprets the evidence and records the next step.

LIVE DEMONSTRATION FLOW

Two cases: documented closure, then an open/escalated investigation

0:00–0:35	**Case A — documented closure**	From Queue, open Evidence only. Show why the closed outcome has recorded support; do not alter it.
0:35–1:15	**Case B — open/escalated**	Open the default review case; distinguish Status, Rule(s), and Action.
1:15–2:05	**Workspace**	Run saved/calculated review; show validator and evidence-confidence limitations.
2:05–2:55	**Evidence & RAG**	Open Source & chunks; explain FTS5/BM25 lexical retrieval—not vectors. Admin view is optional if already unlocked; never show the passcode.
2:55–3:40	**Ask InvestigateIQ**	Ask one bounded case question; show the response, citations, and limits.
3:40–4:50	**Human decision / audit**	Show Close, Request more information, and Escalate options plus required rationale; do not submit or close any case.

Measure the benefit before claiming ROI

Efficiency Verified minutes saved locating case records and source passages

Rework Avoided evidence requests and Compliance returns

Audit readiness Time to reconstruct evidence, rationale, and outcome

Safety Unsupported claims, broken citations, unclear states found in review

ROI % = (verified annual benefit − implementation and run cost) ÷ implementation and run cost × 100

Roadmap: test fixtures + citation evaluation → authentication + lineage versioning → governed ingestion + approved integrations

Planning model only: calculate ROI only after a baseline, approved cost boundary, and verified pilot outcomes.

THANK YOU

Trace the evidence. Own the decision.

Traceable evidence

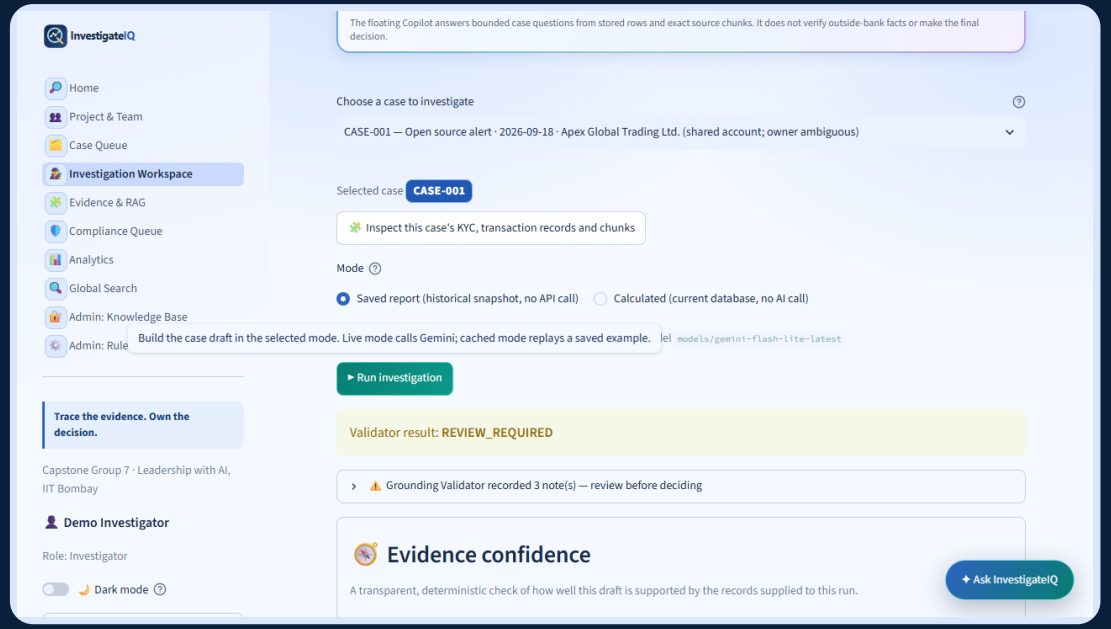
Records, chunks, citations, and visible gaps

Responsible AI

Confidence limitations, not automated judgement

Human accountability

Named rationale and separate Compliance review



Thank you. The FAQ appendix is available for any follow-up question.

FAQ APPENDIX 1

Where does the data come from, and how is it relevant?

What is the data source?

All cases, customers, transactions, documents, and guidance are fictional course material stored locally for this demonstration. The product is not connected to a bank feed or real customer records.

How is retrieval relevant to a case?

The active case/scenario scopes the available evidence. SQLite FTS5 performs lexical keyword matching and BM25 ranks matching passages. The user can open the exact record or chunk behind a result.

Does a citation prove a claim?

No. It is a pointer to inspect supplied text. The product labels evidence gaps and requires a human to interpret the material and record the next action.

FAQ APPENDIX 2

Where is AI used, and how do we reduce hallucination risk?

Where is AI used?

Five specialist components collect/compute evidence. The Investigation Summary Agent is the only component that makes an optional Gemini model call; calculated mode makes no model call.

What guardrails exist?

Prompt rules require cited source IDs, avoid verdict language, and treat source fields as untrusted. The validator checks selected citations, ratios, and unsupported conclusion wording.

Can the system guarantee truth?

No. These checks reduce specific failure modes only. Evidence confidence exposes limitations, and a named human—not the AI—decides whether to close, request information, or escalate.

FAQ APPENDIX 3

How are access, audit, and value handled?

Is this role-based access control?

No. The visible role selector is a demo display label. Admin pages use a limited passcode gate, which is not production authentication or authorization; the code is not shown in the deck or Blueprint.

What is in the audit trail?

System, AI, and human workflow events are recorded with case context. A human decision requires a name and rationale and becomes a recorded state; a Compliance return creates a new auditable step.

How much money or time does it save?

No percentage or financial claim is made from fictional data. A real pilot would measure verified lookup time, rework, audit reconstruction time, and safety issues against an approved cost baseline.